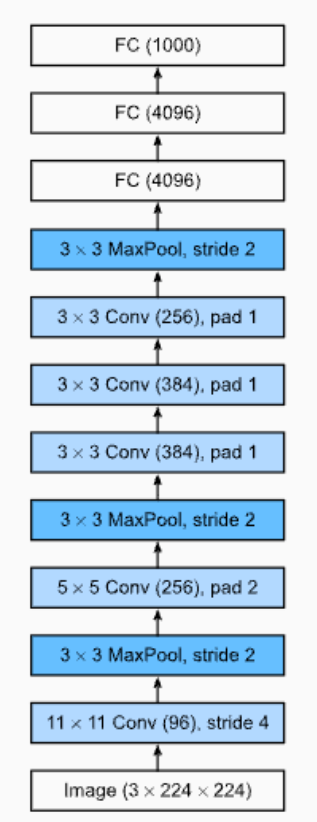
Lab: Image Classification and Feature Visualization Using AlexNet

Objective

Select your own image dataset and use PyTorch to implement AlexNet for image classification. Focus on understanding how CNN layers extract and transform image features, rather than only achieving high accuracy.

Instructions

1. **Select a Dataset:** Choose a public image dataset with at least two classes. Describe the dataset and prepare the images for training, validation, and testing.
2. **Implement AlexNet:** Train AlexNet using PyTorch. You need to implement the AlextNet architecture yourself. Record training and validation results.



3. **Explore the Network:** Print the AlexNet architecture and output tensor shapes at major layers, including convolution, activation, pooling, and fully connected layers.

4. **Visualize Feature Maps:** Select images from your dataset and visualize afew of feature maps from the early, middle, and deep convolutional layers. Compare feature maps from two images belonging to different classes.
5. **Analyze Your Observations:** Explain how feature maps change across layers, how different layers contribute to feature extraction, and how the extracted features help classify images.
6. **Train model:** Train model with 20 epoches and using EarlyStoping technique, save training model after finishing training.
7. **Evaluate the Model:** Load trainined model for final evaluation, report test accuracy, display a confusion matrix, and discuss examples of correct and incorrect predictions.

Use of AI Tools

You may use AI tools to understand concepts, write or debug code, and interpret results. However, you must understand your implementation and explain your own observations. Briefly acknowledge how you used AI tools.

Submission

Submit a Jupyter Notebook (`.ipynb`) containing your code, visualizations, experimental results, and explanations.

Main focus: Understand how AlexNet transforms raw pixels into increasingly complex features and uses them to distinguish between image classes.